

A Medical Data Sharing Scheme Based on Blockchain Attribute-based Searchable Encryption

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Abstract—Aiming at the problems of low sharing efficiency and easy privacy leakage of electronic medical records, a blockchain-based attribute-based searchable encrypted electronic medical record sharing scheme is proposed to achieve fine-grained access control of users' search privileges in one-to-many data sharing scenarios, and at the same time, solve the inefficiency of ciphertext retrieval. In this paper, the shared medical data is divided into structured and unstructured data, and the data is stored in a combined on-chain and off-chain mode, which ensures that both participate in the sharing at the same time, and also relieves the storage pressure of the blockchain. Attribute encryption based on ciphertext policy provides fine-grained access control services, and a data dictionary index structure is designed to complete ciphertext search in the index using searchable encryption. Blockchain nodes retrieve ciphertext keywords and search the index to improve the search efficiency of blockchain nodes. Experimental analysis shows that the scheme can quickly perform the search and sharing of electronic medical records.

Keywords—component; Blockchain; medical data; access control; searchable encryption; CP-ABE;

I. INTRODUCTION

Traditional centralized cloud based medical data sharing solutions rely on a trusted third party, which may lead to data leakage issues. Additionally, data encryption can protect data privacy, but ciphertext is difficult to regulate and share. Due to the emergence of blockchain technology, solutions based on blockchain have been widely discussed [1]. By combining with encryption technology, the privacy and security of medical data access control systems can be ensured [2], but it faces the challenge of low efficiency in ciphertext search, and due to the openness and accessibility of blockchain, the lack of fine-grained access control is also evident [3].

As a new type of public key encryption system, the ciphertext attribute-based encryption algorithm (CP-ABE) achieves "one to many fine-grained" ciphertext access control. The searchable encryption algorithm allows users to search for keywords in the ciphertext domain without leaking plaintext information. At present, there has been a lot of research in the academic community on the application of attribute-based encryption and searchable encryption in various fields. Reference [4] proposes using blockchain authorization mechanism and ABE encryption, and introducing many to many matching to achieve access control. Reference [5] outsources the encryption and decryption of the CP-ABE algorithm to fog nodes, allowing data owners to effectively manage their strategies. Reference [6] designed a dynamic multi keyword searchable encryption scheme on encrypted cloud files to improve search efficiency. Reference [7] divides data into public and private parts, and uses symmetric searchable encryption to encrypt these two parts separately, which overall improves the efficiency of data sharing. In summary, in order to solve the problems of low efficiency of medical record search and easy privacy leakage when sharing medical data, this paper proposes a trusted medical data sharing scheme that combines attribute-based searchable encryption and blockchain technology. The main contributions are as follows:

- (1) Combining attribute-based encryption algorithms with searchable encryption algorithms to achieve fine-grained access control of data while making it searchable in ciphertext state.
- (2) Divide medical sharing data into structured and unstructured data (such as CT images), encrypt and process them to build an index and store them in the blockchain, and use searchable encryption to achieve secure retrieval.

(3) Build a data dictionary index structure, which consists of two parts: file index and keyword index. The data requester retrieves both structured and unstructured data based on the search trapdoor.

II. PROJECT DESIGN

A. Symbol Explanation

The symbols and their corresponding meanings involved in this article are shown in Table I.

TABLE I. MAIN PARAMETER DESCRIPTION

Symbol	Meaning
DO	Data Owner
DU	Data Requester
KGC	Key Management Center
IPFS	Interstellar File System
PK	System master public key
MSK	System master key
SK	User attribute private key
Uid	User identification
Key	SM4 encryption key
M	Structured data plaintext
M_p	Unstructured data (Image type)
S	User attribute set
T	Access Control Tree
I_w	Keyword Index
W	Keywords extracted from plaintext files
W'	Keywords queried by users
T_q	Search for trapdoors
CT_f	Data ciphertext, SM4 encrypted plaintext data
CT_k	Key ciphertext, CP-ABE algorithm encrypts SM4 key
hashpicture	Shared image data hash address
hash	Key ciphertext hash address
CT_p	Image ciphertext, SM4 encrypted hashpicture value

B. System Overview

The improved attribute-based encryption scheme system architecture in this article consists of five parts, namely data owner DO, data requester DU, key management center KGC, blockchain network, and interstellar file system IPFS. As shown in Fig. 1.

DO is responsible for setting data access policies, data encryption, uploading to IPFS, and index construction; When accessing data, DU matches the generated search trapdoor with the keyword index ciphertext, and the attributes meet the access policy before the encrypted file can be effectively shared; KGC is responsible for initializing system parameters and generating user attribute private keys; The blockchain is responsible for storing user identifiers and indexes, and returning the searched medical data information to the DO; IPFS are responsible for storing CP-ABE encrypted ciphertext and shared image data, and returning IPFS hash addresses.

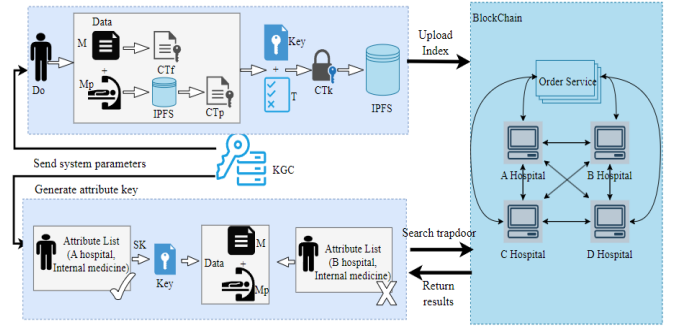


Figure 1. System model diagram.

C. Construction of Data Dictionary Index

The core of this searchable solution is to establish a data dictionary index to achieve fast retrieval of ciphertext. The data dictionary index consists of two parts: file index and keyword index, as shown in Fig. 2.

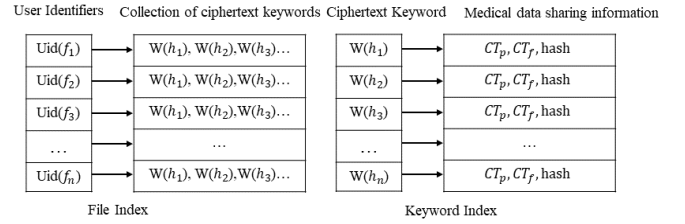


Figure 2. Data Dictionary Index.

The file index is used to store the mapping between the user identifier and all medical data. The key is the user identifier $Uid(f)$, and the value is the ciphertext set of the user's keywords $c=\{W(h1), W(h2), W(h3) \dots\}$. The keyword index key is the ciphertext of the unique keyword for medical data, and the value is the collection of medical data that includes the keyword. The pseudocode for constructing the data dictionary index algorithm is shown in Table II.

TABLE II. CONSTRUCTING INDEX ALGORITHM PSEUDOCODE

Algorithm 1:

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Input: User identifier Uid, keyword to be searched;
Output: Query result Sharedata;
1 keywordciphertext=Encryextractkeywords(M);
2  $I_w = [\text{keywordciphertext}:\text{Sharedata}(CT_f, \text{hash}, CT_p)];$ 
3 KeywordIndex=  $I_w$ ;
4 FileIndex = DataMerge{Uid: Keywordcollection(  $I_w$  )};
5 UploadBC data = AddIndex (FileIndex);
6 set  $T_q = \text{Generate-TrapToken}(W')$ ;
7 if Uid exists in FileIndex then
8   if  $T_q$  exists in KeywordIndex and unique then
9     return Sharedata(  $CT_f$ , hash,  $CT_p$  );
10  else
11    return null;
12 else
13  return null;
```

D. Algorithm Overview

The algorithm involved in this article mainly consists of ten parts: Setup, KeyGen, EncryptP, EncryptM, UpIPFS, IwGen, UploadBC, Trapdoor, Search, and Decrypt.

$$Setup() \rightarrow PK, MSK \quad (1)$$

The Setup algorithm is executed by a trusted management agency.

$$KeyGen(MSK, S) \rightarrow SK \quad (2)$$

The KeyGen algorithm executes the key generation algorithm, inputs MSK and user-defined attribute set S, and outputs the user attribute private key SK.

$$EncryptP(M_p, hashpicture, Key) \rightarrow CT_p \quad (3)$$

The EncryptP algorithm is executed by the data owner, uploads the image to IPFS, returns the hash value hashpicture, and performs SM4 encryption on the hashpicture value.

$$EncryptM(M, Key, PK, T) \rightarrow CT_f, CT_k \quad (4)$$

The EncrypM algorithm takes structured data plaintext M, Key, and access control tree T as inputs, encrypts the Key, and outputs ciphertext CT_f and CT_k .

$$UpIPFS(CT_k) \rightarrow hash \quad (5)$$

The UpIPFS algorithm will CT_k Upload to IPFS for storage and return the hash value of the storage address.

$$IwGen(W, CT_p, CT_f, hash) \rightarrow I_w \quad (6)$$

IwGen algorithm inputs image ciphertext CT_p , data ciphertext CT_f , key ciphertext hash, output keyword index.

$$UploadBC(I_w) \quad (7)$$

UploadBC algorithm index I_w is published to the blockchain for storage and further processing.

$$Trapdoor(PK, W') \rightarrow T_q \quad (8)$$

The Trapdoor algorithm is executed by the data user, queries the keyword W' , and outputs the search trapdoor T_q .

$$Search(T_q, I_w) \rightarrow CT_p, CT_f, hash \quad (9)$$

Search algorithm blockchain node execution, input search trapdoor T_q and keyword index ciphertext I_w , call smart contract matching return $CT_p, CT_f, hash$.

$$Decrypt(PK, CT_p, CT_f, hash, SK) \rightarrow Key, M, M_p \quad (10)$$

The Decrypt algorithm only allows users who meet the attribute policy to successfully decrypt and obtain the Key, thereby obtaining message data plaintext M and image data M_p .

III. EXPERIMENTAL SIMULATION AND ANALYSIS

A. Experimental Environment

In order to verify the feasibility of this scheme, a prototype system was built in this paper for validation. A blockchain network was built using the Hyperledger Fabric 2.2.0 blockchain platform. The software experimental environment used in this article is VMware Workstation, with X64 Ubuntu 20.04 TLS operating system, 2GB of memory, and 120GB of storage.

B. Function Description

The main information shared in this article is medical data, including structured and unstructured data. To conduct performance testing experiments on the shared solution, this article collected medical data. The structured data includes general information, chief complaints, and other textual categories. Unstructured data refers to images such as CT scans, as shown in Fig. 3.

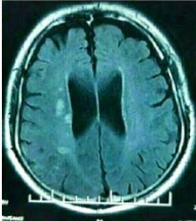
Example medical record		Example CT image
Variable	Content	
General Information	Male, 44 years old	
Chief complaint	Headache for a day.	
Past medical history	No infectious disease history.	
Personal history	Born locally, no smoking.	
family history	nothing	
differential diagnosis	Atherosclerosis.	
clinical diagnosis	cerebral infarction	

Figure 3. Example of shared medical data.

C. Performance Experiment Plan and Results

1) *Encryption and Decryption Stage*: The encryption and decryption efficiency of the CP-ABE algorithm used in this article is limited by the number of attributes. After testing 50 times, take the average and repeat 10 times to calculate the average again. This article uses structured data testing as an example to test two schemes: Scheme one, directly using the CP-ABE algorithm to encrypt and decrypt files; Option 2 of this article proposes encrypting the source file using the SM4 algorithm first, and then encrypting and decrypting the SM4 key using the CP-ABE algorithm, as shown in Fig. 4.

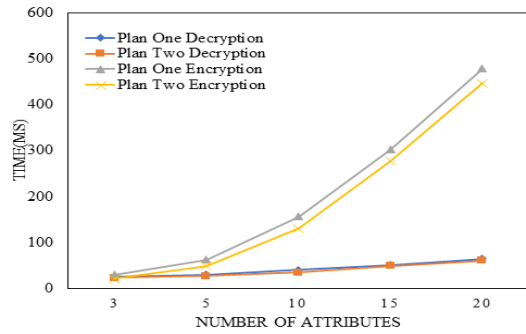


Figure 4. Comparison of encryption and decryption time.

2) *Searchable encryption stage*: The experiment tests the time cost performance of trapdoor generation and search matching by changing the number of keywords. The time for trapdoor and index matching is measured at 100, 200, 300, 400, 500, and 600 keywords. From Fig. 5, it can be seen that as the number of keywords increases, the corresponding execution time of both algorithms slightly increases, while the increase in index matching test is relatively small.

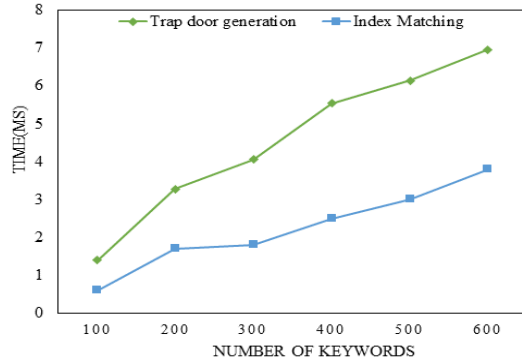


Figure 5. Trapdoor generation and index matching test.

D. Comparative Experiment

In order to effectively analyze the performance of the proposed solution, several similar solutions were selected for comparison with the proposed solution. As shown in Table III.

TABLE III. LITERATURE COMPARISON

Scheme	Keyword Search	Blockchain Technology	Privacy Protection	Access Control
[8]	NO	YES	YES	Strong
[9]	YES	NO	YES	-
This article's proposal	YES	YES	YES	Strong

Reference [8] introduces CP-ABE technology to achieve fine-grained access control of blockchain data, and has built-in encryption technology to ensure data confidentiality, but has weak scalability. The ciphertext and index information of the

scheme in [9] are stored in cloud servers, and the index information is prone to tampering.

IV. CONCLUSIONS

To improve the security and efficiency of data sharing, this paper proposes a blockchain based attribute based searchable encryption data access control scheme. The use of attribute based searchable encryption technology enables fine-grained access control over user search permissions, while establishing a data dictionary index to achieve retrieval in encrypted data state. Blockchain is used to store indexes of medical data, solving the problem of insufficient security in centralized mode. At the same time, the introduction of IPFS storage platform and the use of a combination of on chain and off chain storage mode to store data have alleviated the storage pressure of blockchain. Finally, experiments have shown that the model proposed in this paper has significant advantages and good performance in saving encryption and decryption time and improving ciphertext retrieval efficiency.

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